INCH-POUND
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SUPERSEDING
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PERFORMANCE SPECIFICATION

GASKETS, SYNTHETIC RUBBER, OIL RESISTANT, SLIDE VALVE (FOR 21-INCH SUBMERGED TORPEDO TUBES)

Reactivated after 15 June 2009 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers submarine slide valve gaskets, using oil resistant synthetic rubber.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-601	-	Rubber: Sampling and Testing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-407	-	Visual Inspection Guide for Rubber Molded Items

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D395	-	Standard Test Methods for Rubber Property – Compression Set (DoD adopted)
ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension (DoD adopted)
ASTM D792	-	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement (DoD adopted)
ASTM D2240	-	Standard Test Methods for Rubber Property – Durometer Hardness (DoD adopted)

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 3.1.1 and 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.1 First article inspection. First article inspection shall be performed when one or more of the following apply:

a. Upon initial offering of a material to the government by the vendor.

b. When any change in material formulation is made from previous first article test samples.

c. When any change in manufacturing processes or conditions are made from previous first article test samples.

d. When the product being offered is manufactured at a plant different than that of previous first article test samples.

e. When required by the Naval Sea Systems Command.

3.2 General requirements.

3.2.1 First article samples. Unless otherwise specified (see 6.2), first article samples of the gaskets shall be manufactured using the methods and procedures proposed for the production lot. These samples shall be tested as specified herein, and are for the purpose of determining that the production item meets the requirements of this specification and the design.

3.2.2 Material. The material shall be an oil-resistant synthetic rubber compound of the acrylonitrile-butadiene copolymer type (see 6.3).

3.3 Dimensions. The gaskets shall be furnished in the configuration and to the dimensions on the drawings specified (see 6.2).

3.4 Tolerances. Unless otherwise specified (see 6.2), tolerances shall be held within the limits specified on the drawings.

3.5 Physical requirements. The synthetic rubber material shall conform to the following physical requirements.

3.5.1 Before aging.

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3.5.1.1 Initial tensile strength. Tensile strength shall be 2,500 pounds per square inch (psi), minimum.

3.5.1.2 Initial tensile stress (modulus). Tensile stress at 100 percent elongation shall be as determined.

3.5.1.3 Initial ultimate elongation. Ultimate elongation shall be 250 percent, minimum.

3.5.1.4 Initial hardness. Hardness shall be 75±5 Type A durometer.

3.5.2 After-oven aging. The synthetic rubber material, after having been subjected to the aging test specified in 4.4.7, shall meet the following requirements.

3.5.2.1 Tensile strength. The tensile strength shall be not less than 80 percent of the tensile strength of unaged specimens.

3.5.2.2 Tensile stress. Tensile stress at 100 percent elongation of aged specimens shall be as determined.

3.5.2.3 Ultimate elongation. The ultimate elongation shall be not less than 60 percent of the ultimate elongation of unaged specimens.

3.5.2.4 Compression set. The compression set determined by the method specified in 4.4.7.2 shall be not more then 40 percent.

3.5.2.55 Hardness. The hardness determined by the method specified in 4.4.5 shall be 75±5, Type A durometer.

3.5.3 Oil immersion. After being subjected to the conditions specified in 4.4.8, the synthetic rubber material shall meet the following requirements.

3.5.3.1 Swell. The change in volume of the specimen shall be not more than 3 percent when tested as specified in 4.4.8.1.

3.5.3.2 Tensile strength. The tensile strength shall be not less than 2,000 psi when tested as specified in 4.4.8.2.

3.5.3.3 Tensile stress. Tensile stress at 100 percent elongation when tested as specified in 4.4.8.2 shall be as determined.

3.5.3.4 Ultimate elongation. The ultimate elongation shall be not less than 200 percent when tested as specified in 4.4.8.2.

3.6 Specify gravity. The specific gravity of the synthetic rubber material shall be 1.25±0.10 when determined as specified in 4.4.6.

3.7 Marking. The gaskets shall be marked in accordance with the drawings as specified (see 6.2). In addition, unit packages shall be marked in accordance with the requirements of MIL-STD-129. The nomenclature shall be the exact nomenclature for the part or item that has been approved under the contract order.

3.8 Workmanship. The gaskets shall be molded in a manner that will assure compliance with requirements of this specification. The mold flash shall not exceed 0.003 inch in thickness at the point of contact with the gasket. The gaskets shall be free of blisters, cracks, folds, foreign material, pinholes, tears, wrinkles, or other imperfections that affect the serviceability of the gaskets. The gaskets shall not exhibit any knit lines.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article. First article inspection shall be performed on the gaskets when a first article sample is required (see 3.1). This inspection shall include the examinations and tests in table I.

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TABLE I. First article examinations and tests.

Factor Evaluated	Requirement	Examination or Test
Examinations		
Dimensions	3.3	4.4.1
Tolerances	3.4	4.4.1
Marking	3.7	4.4.2
Workmanship	3.8	4.4.3
Initial Properties		
Tensile strength	3.5.1.1	4.4.4
Tensile stress (modulus)	3.5.1.2	4.4.4
Ultimate elongation	3.5.1.3	4.4.4
Hardness	3.5.1.4	4.4.5
Specific gravity	3.6	4.4.6
Properties After Oven Aging		
Tensile strength	3.5.2.1	4.4.7
Tensile stress	3.5.2.2	4.4.7
Ultimate elongation	3.5.2.3	4.4.7
Compression set	3.5.2.4	4.4.7 and 4.4.7.2
Hardness	3.5.2.5	4.4.7
Properties After Oil immersion		
Swell	3.5.3.1	4.4.8 and 4.4.8.1
Tensile strength	3.5.3.2	4.4.8 and 4.4.8.2
Tensile stress	3.5.3.3	4.4.8 and 4.4.8.2
Ultimate elongation	3.5.3.4	4.4.8 and 4.4.8.2

4.2.1 First article samples and tests. The gaskets and sheets submitted for first article examinations and tests shall be certified to have the same composition, construction, and manufacturing process as the material to be supplied on subsequent lots. When specified (see 6.2), the supplier shall prepare a first article test report. Production of the gaskets by the supplier prior to approval of the procuring activity or the completion of first article examinations and tests shall be at the supplier's risk.

4.2.2 Samples for examinations. A representative sample of five gaskets shall be manufactured using the methods and procedures proposed for production. These shall be evaluated for conformity to the requirements of the contract, specifications, drawings, and the examinations in table I by an activity approved by the procuring activity. Gaskets from batches passing first article tests are suitable for delivery.

4.2.3 Samples for physical property tests other than compression set. Sufficient sheets, 6 inches by 6 inches in plan and 0.080±0.010 inch thick, shall be molded from first article sample stock with an equivalent cure. The sheets shall be tested as specified in table I for initial properties, properties after oven aging (except compression set), and properties after oil immersion. Samples shall be cut from these sheets as required, including three swell samples nominally 1 inch by 2 inches in plan dimensions.

4.2.4 Samples for compression set test. Three compression set samples shall be cut from molded rubber. The rubber shall be molded from first article sample stock with an equivalent cure. The samples shall be in accordance with ASTM D395 Type 1 specimens.

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4.2.5 Rejection for nonconformance. If any of the first article examinations or tests do not conform to the applicable requirements, this shall be cause for denial to proceed with the production of gaskets.

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests for batch acceptance (see 4.3.1) and lot acceptance (see 4.3.2) as specified in table II.

TABLE II. Conformance examinations and tests.

Factor Evaluated	Requirement	Examination or Test
Batch Acceptance (Initial Properties)		
Tensile strength	3.5.1.1	4.4.4
Tensile stress	3.5.1.2	4.4.4
Ultimate elongation	3.5.1.3	4.4.4
Hardness	3.5.1.4	4.4.5
Specific gravity	3.6	4.4.6
Lot Acceptance		
Dimensions	3.3	4.4.1
Tolerances	3.4	4.4.1
Marking	3.7	4.4.2
Workmanship	3.8	4.4.3

4.3.1 Batch acceptance.

4.3.1.1 Batch. A batch consists of a quantity of rubber compound which is mixed at one time and which is to be used in the manufacture of the gaskets. If the curing agents are left out for processing purposes, the batch is referred to as a master batch. It may be subdivided as needed and curing agents incorporated at that time. Each subdivision shall be considered a batch when curing agents are incorporated.

4.3.1.2 Samples for batch tests. Sufficient sheets nominally 6 inches by 6 inches in plan and 0.080±0.010 inch thick shall be molded from each batch or subdivision of master batch with an equivalent cure and shall be tested for the initial properties listed in table II. Three samples shall be tested from each batch or subdivision of a master batch for initial tensile strength, tensile stress, ultimate elongation, hardness, and specific gravity.

4.3.1.3 Rejection of batch. If the sheets in 4.3.1.2 do not conform to all of the requirements for initial properties in table II when tested as specified, the batch from which it was molded shall be rejected and not used for the production of gaskets.

4.3.2 Lot acceptance.

4.3.2.1 Lot. A lot shall consist of gaskets produced in one plant under essentially the same conditions and offered for acceptance at one time. In each case, the number of gaskets offered for inspection constitutes the lot size. For purposes of sampling, a lot shall consist of not more than 2,500 pounds of gaskets of the same design and dimensions produced in one plant under essentially the same conditions.

4.3.2.2 Sampling for lot acceptance. Random samples from each lot of gaskets shall be selected in accordance with table III for lot acceptance examinations.

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TABLE III. Sampling procedure.

Number of Gaskets in Lot	Number of Gaskets to be Selected
1 to 25	All
26 to 50	22
51 to 100	27
101 to 400	33
Over 400	36

4.3.2.3 Rejection of lot. Each of the samples selected in accordance with 4.3.2.2 shall be subjected to the lot acceptance examinations in table II. If any gaskets do not conform to the requirements for lot acceptance in table II, the lot which the gaskets represent shall be rejected. Rejected lots may be submitted for reinspection only after all gaskets are reexamined and all defective gaskets are removed.

4.4 Test methods.

4.4.1 Dimensions and tolerances. Each gasket shall be inspected to ensure that dimension and tolerance requirements are met. All dimensions shall be measured with an instrument capable of measuring to at least one decimal place more than the specified tolerance or, if no tolerance is specified, at least one decimal place more than the specified dimension.

4.4.2 Marking. Each gasket shall be inspected to ensure that gasket marking requirements are met (see 3.7).

4.4.3 Workmanship. Each gasket shall be visually inspected to ensure that gasket workmanship requirements are met. Minor surface defects, including minor defects specified in MIL-STD-407, that do not affect the serviceability of the material shall not be cause for rejection.

4.4.4 Tensile strength, tensile stress, and ultimate elongation. Tensile strength, tensile stress at 100 percent elongation, and ultimate elongation shall be determined in accordance with ASTM D412, Test Method A, with Die C test specimens.

4.4.5 Hardness. Hardness shall be determined in accordance with ASTM D2240, Type A durometer.

4.4.6 Specific gravity. Specific gravity shall be determined in accordance with ASTM D792.

4.4.7 Aging. Specimens for tensile strength, tensile stress, ultimate elongation, hardness, and compression set tests following aging shall be aged in a circulating air oven at a temperature of 194±2 °F (90±1.1 °C) for 46 hours. The aged samples shall be tested not less than 20 hours nor more than 48 hours after removal from the oven. After aging, specimens shall be tested in accordance with 4.4.7.1, 4.4.7.2, and 4.4.7.3.

4.4.7.1 Tensile strength, tensile stress, and ultimate elongation after aging. Tensile strength, tensile stress, and ultimate elongation shall be determined in accordance with 4.4.4 after the tensile specimens have been aged in accordance with 4.4.7.

4.4.7.2 Compression set after aging. Three specimens shall be tested in accordance with ASTM D395, Test Method B, using a 25 percent deflection after being aged in accordance with 4.4.7.

4.4.7.3 Hardness after aging. Hardness shall be determined in accordance with 4.4.5 after the hardness specimens have been aged in accordance with 4.4.7.

4.4.8 Oil immersion. Specimens for swell, tensile strength, tensile stress, and ultimate elongation tests following oil immersion shall be immersed for 94 hours at 158±3.6 °F in a petroleum-base oil having the same viscosity, aniline point, and flash point as the No. 3 oil of Method 6001 of FED-STD-601. After oil immersion, specimens shall be tested in accordance with 4.4.8.1 and 4.4.8.2.

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4.4.8.1 Swell. The volume of each specimen shall be measured in accordance with Method 6231 of FED-STD-601 after immersion in accordance with 4.4.8. Swell shall be based on the average volume change of the three specimens.

4.4.8.2 Tensile strength, tensile stress, and ultimate elongation after oil immersion. Three specimens shall be tested for tensile strength, tensile stress, and ultimate elongation after being immersed in oil in accordance with 4.4.8 except that the specimens shall be tested within 60 minutes from the time of removal from the immersion fluid. Tensile strength and tensile stress shall be determined on the basis of the swollen area as described in Method 6111 of FED-STD-601.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gaskets covered by this specification form the hydrostatic seal in the slide valve in hydraulic ejection submarine torpedo tubes. This valve controls the flow of sea water between the torpedo ejection pump and the individual torpedo tubes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, when required (see 3.1 and 3.1.1).

c. Dimensions and tolerances of material required (see 3.3 and 3.4).

d. Marking required (see 3.7).

e. First article test report (see 4.2.1).

f. Packaging requirements (see 5.1).

6.3 Optional compound. Requirements of this specification have been met by a compound of the following formula:

Hycar 1042……	100.00	Sulfur……..	1.50
Zinc oxide…….	5.00	Tricresyl phosphate……	10.00
Spheron 9……..	80.67	Dibutyl phthalate……	10.00
Cumar MH-2½……..	6.66	Captax (MBT)……	1.00
Agerite powder…….	1.00	Tetramethylthiuram disulfide….	0.67
Steric acid…….	1.50		

6.4 Description. The gaskets to which this specification applies form the hydrostatic seal in the slide valve in hydraulic ejection submarine torpedo tubes. This valve is described in NAVSEA OP 1933, "21-Inch Submerged Torpedo Tubes Mark 43 and 45 and Mods Description, Operation and Maintenance", and controls the flow of seawater between the torpedo ejection pump and the individual torpedo tubes.

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6.5 Subject term (key word) listing.

Copolymer

Elastomer

Nitrile

Polymer

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Preparing Activity:
Navy – SH
(Project 5330-2008-006)

Review Activities:
Navy – OS
DLA – IS

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